

VL 312: Electronics Systems Packaging (ESP)

Course Code	VL 312
Course Name	Electronics Systems Packaging (ESP)
L-T-P-C	3-0-0-3
Prerequisites	3rd Sem VL 201/EE 311, FE for EE & GE, Core for VLSI
Intended for	Discipline Core for B.Tech. in VLSI and Microelectronics Engineering students
Mutual Exclusion	None
Approval	61st BoA (Jan 2026); 49th Senate (Jun 2026), in the Minor in VLSI Technology curriculum

Preamble

Packaging is crucial for the performance of Electronics & Microsystems. It not only encases components but also connects the die to the system, protects against environmental threats, enables efficient heat transfer, and ensures reliable power and signal delivery. As technology advances, the need for cost-effective, space-efficient, high-speed, and reliable packaging grows. Professionals in this field must understand advanced technologies to create high-performance packages affordably. This course provides an opportunity for students to explore the latest packaging technologies and their interconnections, ultimately enhancing performance, manufacturability, functionality, and reliability in electronics & microelectronics. Join us to help shape the future of microelectronic systems.

Course Modules

- **Unit-1: Introduction to Electronic System Packaging:** Introduction to electronic systems and the significance of packaging, Historical perspective and evolution of semiconductor packaging, Different levels of packaging: chip-level, board-level, and system-level. (8 hours)
- **Unit-2: Chip Processing and Electronic Packaging:** Basics of semiconductor manufacturing processes, Wafer fabrication, inspection, and testing procedures. Evolution of wafer packaging and chip connection techniques, including wire bonding, Tape Automated Bonding (TAB), and flip-chip methods, Package materials, thermal mismatch in packages, and current packaging trends and various applications. (8 hours)
- **Unit-3: Integrated Packaging & Computer-Aided Design (CAD):** Electrical, Thermal, and Mechanical Perspective of Resistive Parasitic, Capacitive Parasitic, and Inductive Parasitic, Signal integrity and power distribution challenges of electrical, thermal, and mechanical design. Introduction to Computer-Aided Design (CAD) tools for PCB design. Design flow considerations, including schematic capture and component layout. Design for Manufacturability (DFM), Reliability (DFR), and Testability (DFT). (8 hours)
- **Unit-4: Materials Consideration & Process Flow for Packaging:** Materials properties for packaging, Selection criteria based on application requirements. Printed circuit wiring & boards (PWB & PCBs), flexible circuits, Process flow-chart PCB and PWD substrates fabrication. Surface Mount Technology (SMT): Benefits and design considerations of SMT. Soldering techniques: reflow wave soldering, lead-free soldering, and green electronics. Process flow: photoresist and application methods, UV exposure and developing PWB etching, Resist stripping, Screen-printing technology, Through-hole manufacture process steps, and pattern plating methods. (8 hours)
- **Unit-5: Advanced Packaging Technologies & Reliability:** Assessment of package reliability under various conditions, Failure mechanisms and testing methodologies, Heterogeneous integration and 3D packaging, Emerging trends in electronics packaging. (8 hours)

Text Books

1. *Fundamentals of Microsystems Packaging* (Ed. Rao Tummala), McGraw-Hill, 2001

References

1. *Advanced Electronic Packaging*, Richard K. Ulrich and William D. Brown, Second Edition, IEEE Press Series on Microelectronic Systems, A Johan Wiley & Sons, INC. Publication, 2006.
2. *Microelectronics Packaging Handbook*, Tummala, R.R., 3 volumes, Chapman Hall, 1996.